Survive & Endure

Software Engineering COMP 3040
University of Massachusetts Lowell Fall 2025

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Project Overview

- **Endure and Survive** is a *2D educational survival* game
- Designed for *students* in grades *4–6*
- Players learn *survival skills* through *interactive* gameplay
- Focused on *fire building*, *shelter construction*, and *basic navigation*

Project Goals

Problem

- Survival skills are difficult to teach <u>safely</u> in a classroom setting.
- Students often learn better through hands-on practice, rather than lectures.
- This is important because improper understanding of survival skills can lead to <u>real world</u> risks.

Methodology

- We use a 2D educational game to *simulate* real survival scenarios.
- Students may attempt survival situations without the real world consequences of failure.

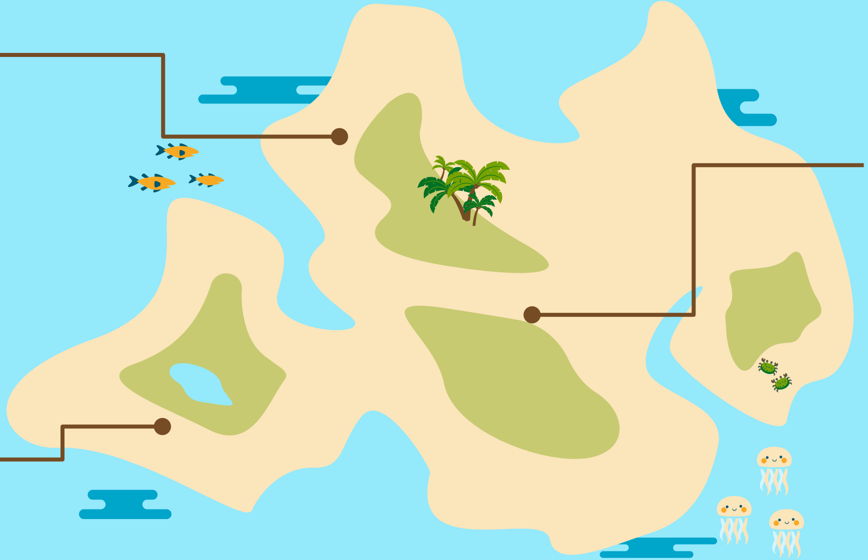
Game Levels

Starting a Fire

Player learns essential ingredients for fire: **tinder**, **kindling**, and **fuel**.

Building Shelter

Player collects logs to construct a **lean-to** shelter.



Orienteering

Player uses basic **compass** reading skills to find key locations.

Domain Research 🏝️

- Researched beginner-level camping and survival standards.
- Researched existing educational games for *engagement* and *learning* balance.
- Focused on helping create an accurate and safe understanding of survival techniques.

Project Constraints 🏝️

- Avoid unsafe or misleading survival instructions

- Game instructions must be written so they can easily be understood by 4th – 6th graders

- Must remain educational while being fun and engaging to play.

Technologies Used 🏝️

Godot

- Used to develop the 2D game and manage gameplay logic.

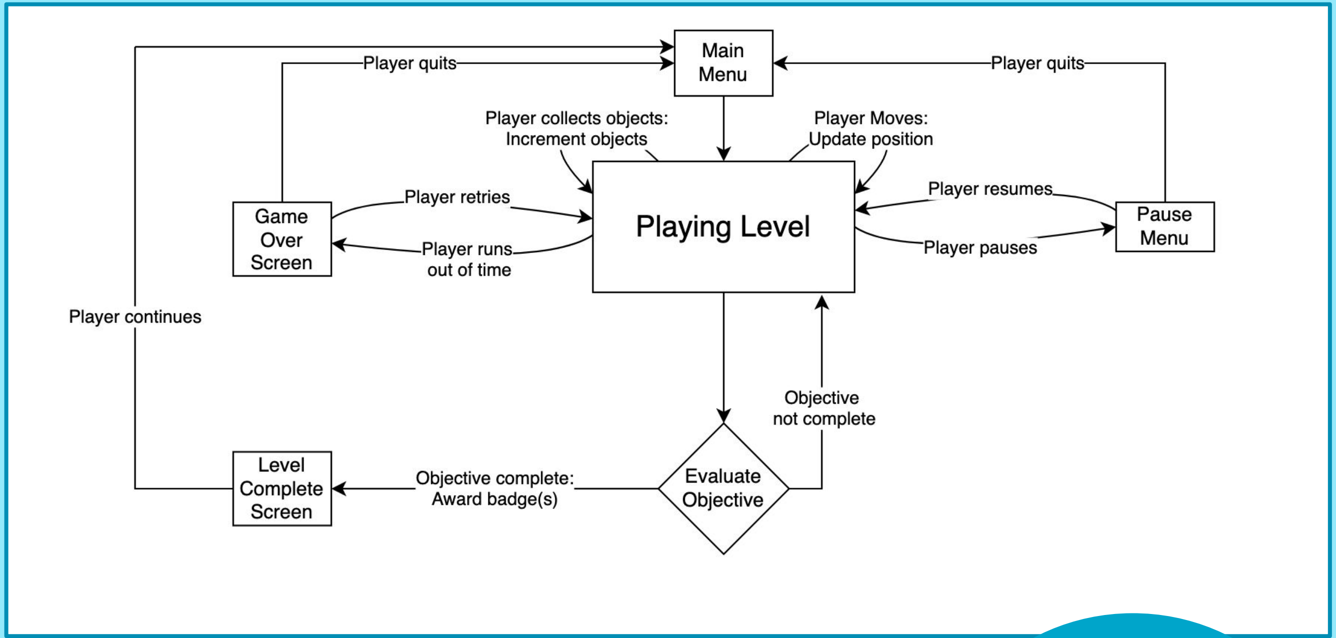
Git & GitHub

- Used for version control and team collaboration.
- Allows multiple developers to work simultaneously without overwriting code.

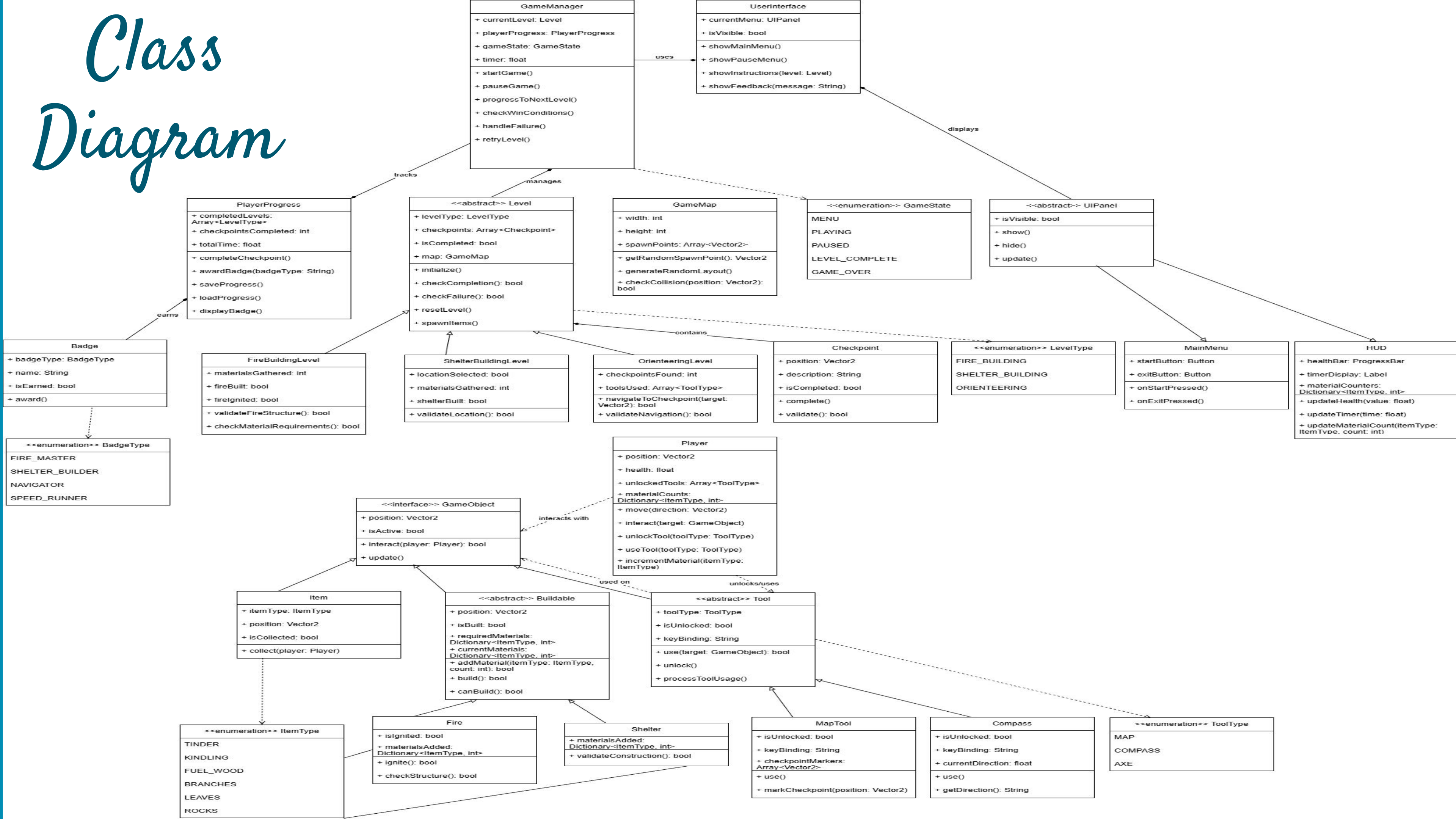
GODOT
Game engine

State Diagram

- Primarily transitioning between *Playing Level* and various menus
- Basic logic for all levels is the same: Move around and complete assigned tasks
- Only diagram that reasonably fits in the presentation



Class Diagram



Demonstration

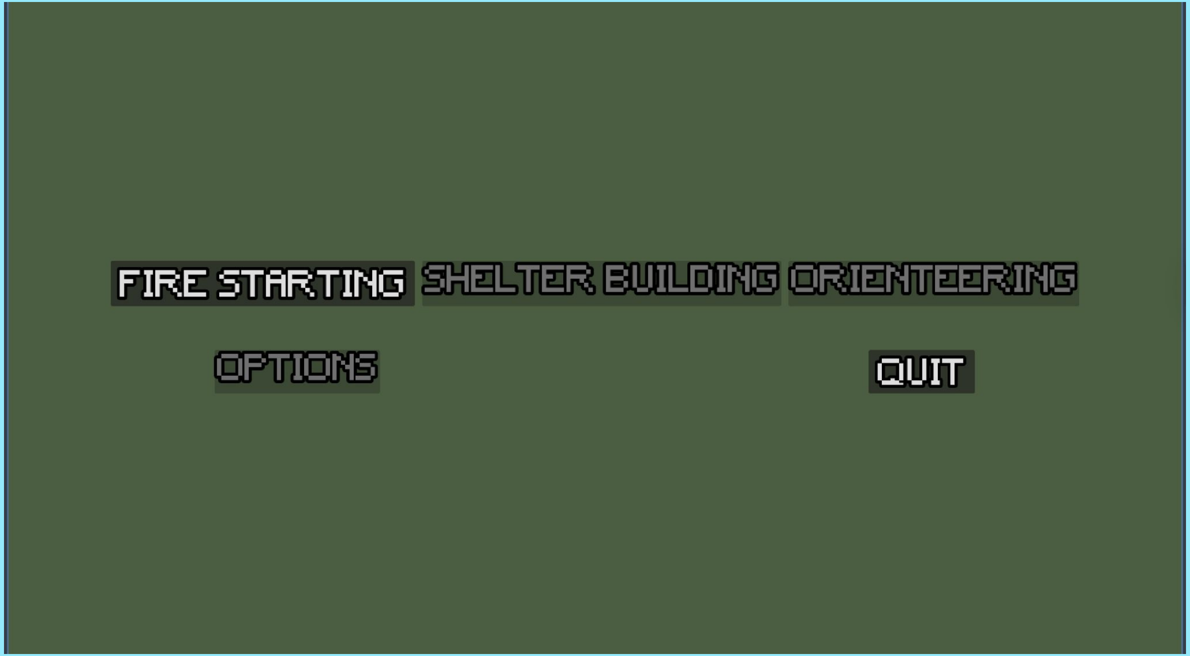
Interface

- Top-down 2D island map
- Simple keyboard and mouse controls
- On-screen instructions and timers
- Visual feedback for progress and errors

Scene 1

Main Menu

- Select level, change settings*, or exit
- First level begins unlocked
  - Additional levels unlock after the previous is beaten



*not implemented

Scene 2

Level 1

1. Players explore while collecting tinder, kindling, and fuel
2. Once all materials are collected, players must find a suitable fire location
3. Minigame simulates* constructing a fire - Click-&-drag materials in order: ***tinder* → *kindling* → *fuel***

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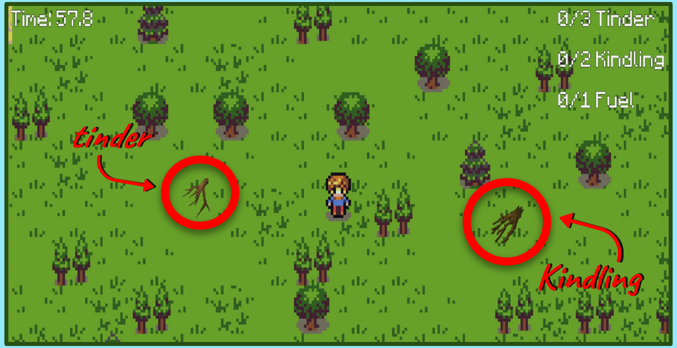

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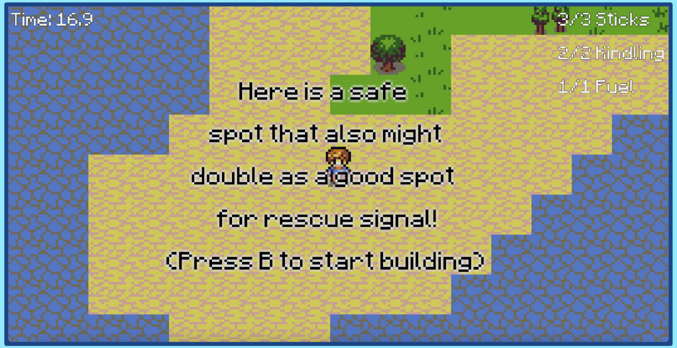

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*Drag-&-drop "simulation" is a placeholder for a more in-depth

Scene 3

Level 2

- Similar to level 1.
- Players must collect 5 logs and find a suitable location for a shelter.
- Minigame simulates constructing a lean-to.

Scene 4

Level 3

- Players must navigate to two locations on the map using cardinal directions.
- Text prompts indicate the direction of the target location, relative to the player.
- A compass rose is provided so the player may orient themselves.

Future Improvements

- **Improve existing systems**
  - More interactive + informative
  - Increased instruction + feedback
- **Create additional systems**
  - Keep gameplay fresh and engaging
  - Teach additional survival skills
    - E.g. weather/terrain variety, difficulty levels
- **Improve UX**
  - SFX/VFX - e.g. fire sound/animation
  - User settings
  - Bug fixes

Thanks for Listening

Any Questions?

Acknowledgements

We gratefully acknowledge and appreciate participation of our customer Dr. Daly from the University of Massachusetts Lowell